



AI Engineer Roadmap 2026

Complete AI Engineer Learning Path with Timeline, Courses, Certifications, Projects & Career Roadmap



Introduction

AI Engineering is one of the fastest-growing careers in technology. Modern AI Engineers focus on building AI-powered applications, integrating LLM APIs, creating AI workflows, and deploying AI SaaS products.



Skills Required

- Python Programming
- Machine Learning Basics
- Deep Learning Fundamentals
- LLM Engineering
- Prompt Engineering
- Vector Databases
- RAG Systems
- AI Agents
- Cloud Deployment
- Docker & DevOps



12-Month AI Engineer Timeline

Month	Focus
1	Python + Git + Linux
2	Mathematics + Statistics
3	Machine Learning
4	Deep Learning
5	LLMs + Prompt Engineering
6	LangChain + RAG
7	Vector Databases
8	AI Agents
9	FastAPI + Backend
10	Docker + Deployment
11	Portfolio Projects
12	Interview Prep + Freelancing

Phase 0 — Developer Foundations

Topics Covered:

- Internet Basics
- Git & GitHub
- Linux Basics
- APIs & JSON

Free Learning Resources:

- <https://skills.github.com/>
- <https://linuxjourney.com/>

Phase 1 — Python Programming

Topics Covered:

- Variables
- Loops
- Functions
- OOP
- NumPy
- Pandas

Free Learning Resources:

- <https://www.py4e.com/>
- <https://www.freecodecamp.org/learn/scientific-computing-with-python/>

Phase 2 — Mathematics for AI

Topics Covered:

- Statistics
- Linear Algebra
- Probability
- Calculus Basics

Free Learning Resources:

- <https://www.khanacademy.org/math/statistics-probability>
- <https://www.3blue1brown.com/topics/linear-algebra>

Phase 3 — Machine Learning

Topics Covered:

- Regression
- Classification
- Decision Trees
- Random Forest

Free Learning Resources:

- <https://www.coursera.org/learn/machine-learning>
- <https://www.kaggle.com/learn/intro-to-machine-learning>

Phase 4 — Deep Learning

Topics Covered:

- Neural Networks
- CNN
- RNN
- Transformers

Free Learning Resources:

- <https://course.fast.ai/>
- <https://pytorch.org/tutorials/>

Phase 5 — Generative AI & LLM Engineering

Topics Covered:

- Prompt Engineering
- RAG
- Embeddings
- AI Agents

Free Learning Resources:

- <https://academy.openai.com/>
- <https://huggingface.co/learn>

Phase 6 — AI App Development

Topics Covered:

- FastAPI
- Docker
- Streamlit
- Deployment

Free Learning Resources:

- <https://fastapi.tiangolo.com/>
- <https://docs.docker.com/get-started/>

Portfolio Projects

- ChatGPT Clone
- AI Resume Screener
- RAG Chatbot
- AI SaaS Platform
- AI Automation System
- Voice AI Assistant

Best Certifications

Certification	Importance
Hugging Face	High
DeepLearning.AI	Excellent
Google AI Certificates	High
Microsoft AI Skills	High
Kaggle Certificates	Good

Recommended Tech Stack

Category	Tools
Language	Python
Backend	FastAPI
Frontend	Next.js
LLM Frameworks	LangChain
Vector DB	Pinecone
Database	PostgreSQL
Deployment	Docker

Interview Preparation

- AI System Design
- RAG Architecture
- Prompt Engineering

- LLM Optimization
- Vector Databases
- API Integration
- AI Deployment

Final Advice

The fastest path to becoming an AI Engineer is to learn Python, build projects consistently, work with LLM APIs, deploy AI applications publicly, and create a strong GitHub portfolio.

Useful Resources

- <https://academy.openai.com/>
- <https://huggingface.co/learn>
- <https://www.deeplearning.ai/>
- <https://www.freecodecamp.org/>